

NUnit Unit Testing for UtilLib - Implementation Document

Introduction

The purpose of this exercise is to learn how to create automated unit tests using the NUnit framework. The UtilLib project contains a method named `ParseHostName()`, which extracts the host name from a given URL. Unit tests are written to verify that the method works correctly under different input conditions.

Objectives

- Learn how to create automated unit tests using NUnit.
- Understand NUnit custom attributes such as `[TestFixture]` and `[Test]`.
- Verify the `ParseHostName()` method with different scenarios.
- Follow the Single Assertion Rule.
- Use `Assert.That()` for validation.

Implementation Steps

Step 1: Create a new Class Library project in the same solution and name it UtilLib.Tests.

Step 2: Rename the default test class to `UrlHostNameParserTests.cs`.

Step 3: Add a reference to the UtilLib project.

Step 4: Install NUnit and NUnit3TestAdapter using NuGet Package Manager.

Step 5: Create test methods following the naming convention `UnitUnderTest_Scenario_ExpectedOutcome`.

Step 6: Write at least two test cases to cover both execution paths of `ParseHostName()`.

Step 7: Use `Assert.That()` with a single assertion in each test method.

Step 8: Build the solution and run all unit tests.

Step 9: Modify the `ParseHostName()` logic intentionally to break one test.

Step 10: Run the tests again and observe the failed test result in Test Explorer.

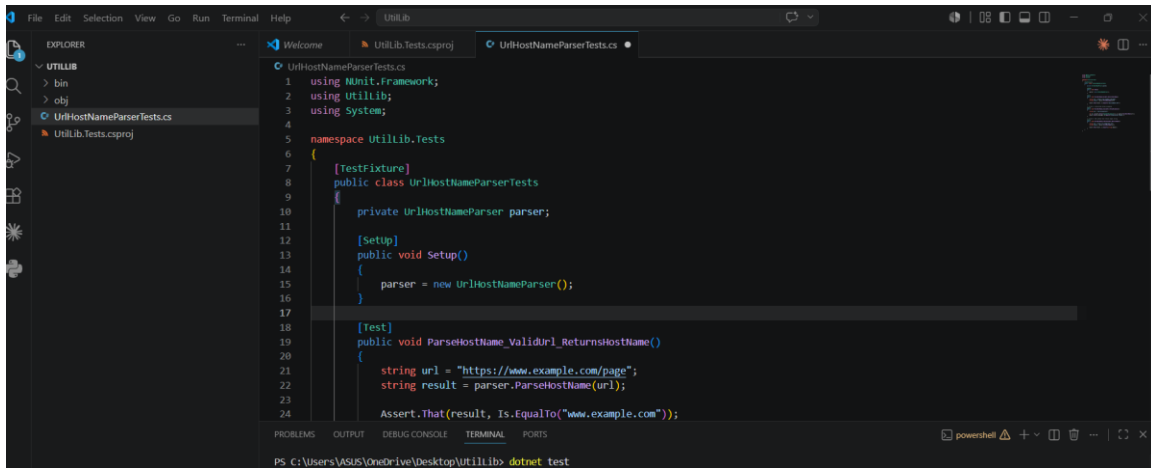
Expected Result

All unit tests should pass when the `ParseHostName()` method is implemented correctly. After modifying the source code intentionally, one or more test cases should fail, proving that the tests correctly validate the functionality.

Conclusion

This exercise helped in understanding how NUnit can be used to automate testing. Using proper test methods, assertions, and naming conventions improves software quality and helps detect errors quickly during development.

Execution



The screenshot shows the Visual Studio IDE with the following details:

- EXPLORER:** The project structure shows 'UTILIB' with subfolders 'bin' and 'obj'. The file 'UtilHostNameParserTests.cs' is selected under 'UTILIB.Tests.csproj'.
- Code Editor:** The file 'UtilHostNameParserTests.cs' is open, showing the following code:

```
1 using NUnit.Framework;
2 using UtilLib;
3 using System;
4
5 namespace UtilLib.Tests
6 {
7     [TestFixture]
8     public class UtilHostNameParserTests
9     {
10         private UtilHostNameParser parser;
11
12         [Setup]
13         public void Setup()
14         {
15             parser = new UtilHostNameParser();
16         }
17
18         [Test]
19         public void ParseHostName_ValidUrl_ReturnsHostName()
20         {
21             string url = "https://www.example.com/page";
22             string result = parser.ParseHostName(url);
23             Assert.That(result, Is.EqualTo("www.example.com"));
24         }
25     }
26 }
```
- TERMINAL:** The terminal shows the command 'PS C:\Users\VASU\OneDrive\Desktop\UtilLib> dotnet test'.